

AE_544_LecNote07__Hamiltonian_Dynamics__Ch07

Disclaimers

This repository contains markdown-based lecture notes for the Spring 2025 semester. These materials are intended solely for students enrolled in the course during this term. Mistakes may exist and will be corrected as they are identified, but the official textbook remains the ultimate reference for accuracy.

We will introduce the canonical form of the equations of motion, and associated canonical coordinate transformations.

Hamiltonian Function

The Hamiltonian function $\mathcal{H}$ is defined as

$$\mathcal{H} \equiv \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \dot{q}_i - \mathcal{L} = \mathcal{H}(\mathbf{q}, \dot{\mathbf{q}}, t)$$

Hamiltonian $\mathcal{H}$ is closely related to the Lagrangian function $\mathcal{L}$.

$$\frac{d\mathcal{H}}{dt} = \sum_{i=1}^n \left[\frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) \dot{q}_i + \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \ddot{q}_i \right] - \left[\sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial q_i} \dot{q}_i + \sum_{i=1}^n \frac{\partial \mathcal{L}}{\partial \dot{q}_i} \ddot{q}_i + \frac{\partial \mathcal{L}}{\partial t} \right]$$

Observing the cancellation of the two terms, and re-collecting, we have

$$\frac{d\mathcal{H}}{dt} = \sum_{i=1}^n \left\{ \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) - \frac{\partial \mathcal{L}}{\partial q_i} \right\} \dot{q}_i - \frac{\partial \mathcal{L}}{\partial t} \quad (7.7)$$

1. The first term will vanish as Lagrange's equations for a conservative or holonomic systems.
2. The second term vanishes when the Lagrangian function $\mathcal{L}$ does not explicitly depend on time, i.e., is scleronomic.

If we defined the generalized momentum p_i as

$$p_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{q}_i}$$

the Hamiltonian function $\mathcal{H}$ can be rewritten as

$$\mathcal{H} = \sum_{i=1}^n p_i \dot{q}_i - \mathcal{L} = \mathbf{p}^T \dot{\mathbf{q}} - \mathcal{L}$$

Note that here $\mathbf{p}$ and $\mathbf{q}$ are treated as ~~coordinates of vectors~~ 3×1 matrices rather than abstract vectors.

The Lagrangian $\mathcal{L}$ has a quadratic dependence on $\dot{\mathbf{q}}$, which means we have the relationship

$$\begin{aligned} \mathbf{p} &= \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}} = [\mathbf{B}(t, \mathbf{q})] \cdot \dot{\mathbf{q}} \\ \dot{\mathbf{q}} &= [\mathbf{B}(t, \mathbf{q})]^{-1} \cdot \mathbf{p} \end{aligned}$$

In other words, $\dot{\mathbf{q}}$ and $\mathbf{p}$ are alternative velocity (or momentum) descriptions of the system motion.

$\mathcal{H}$ is Total Energy in Natural Systems

The Hamiltonian function is frequently found to equal the total energy, but this is not universally true from the definition in Eq. (7.1)

Plug in $\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t) = T(\mathbf{q}, \dot{\mathbf{q}}, t) - V(\mathbf{q}, t)$ into Eq. (7.1), we have

$$\mathcal{H}(\mathbf{q}, \dot{\mathbf{q}}, t) = \sum_{i=1}^n \left(\frac{\partial T(\mathbf{q}, \dot{\mathbf{q}}, t)}{\partial \dot{q}_i} \right) \dot{q}_i - T(\mathbf{q}, \dot{\mathbf{q}}, t) + V(\mathbf{q}, t) \quad (7.8)$$

The most general structure of kinetic energy is

$$T = T_0 + T_1 + T_2$$

where

$$\begin{aligned}
T_0 &= T_0(t, \mathbf{q}) && \text{no dependence on } \dot{\mathbf{q}} \\
T_1 &= T_1(t, \mathbf{q}, \dot{\mathbf{q}}) = \sum_{i=1}^n b_i(t, \mathbf{q}) \dot{q}_i && \text{linear dependence on } \dot{\mathbf{q}} \\
T_2 &= T_2(t, \mathbf{q}, \dot{\mathbf{q}}) = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n m_{ij}(t, \mathbf{q}) \dot{q}_i \dot{q}_j && \text{quadratic dependence on } \dot{\mathbf{q}}
\end{aligned}$$

Note that m_{ij} doesn't have to be mass, for example it will be momentum of inertial for single pendulum, or can be entries of matrix of inertia in case of rigid-body dynamics.

With this decomposition, we have

$$\begin{aligned}
\frac{\partial T}{\partial \dot{q}_i} &= \frac{\partial T_0}{\partial \dot{q}_i} + \frac{\partial T_1}{\partial \dot{q}_i} + \frac{\partial T_2}{\partial \dot{q}_i} \\
&= 0 + b_i(t, \mathbf{q}) + \sum_{j=1}^n m_{ij}(t, \mathbf{q}) \dot{q}_j \text{correct this}
\end{aligned}$$

Plugging it into Eq. (7.8) gives

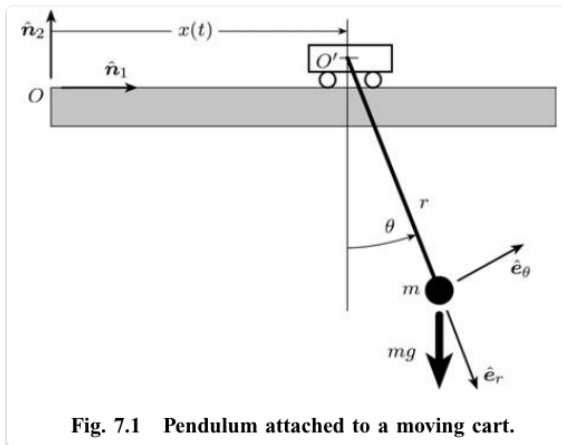
$$\begin{aligned}
\mathcal{H}(\mathbf{q}, \dot{\mathbf{q}}, t) &= \sum_{i=1}^n \left(\frac{\partial T(\mathbf{q}, \dot{\mathbf{q}}, t)}{\partial \dot{q}_i} \right) \dot{q}_i - T(\mathbf{q}, \dot{\mathbf{q}}, t) + V(\mathbf{q}, t) \\
&= \sum_{i=1}^n \left(b_i(t, \mathbf{q}) + \sum_{j=1}^n m_{ij}(t, \mathbf{q}) \dot{q}_j \right) \dot{q}_i - T(\mathbf{q}, \dot{\mathbf{q}}, t) + V(\mathbf{q}, t) \\
&= \sum_{i=1}^n b_i(t, \mathbf{q}) \dot{q}_i + \sum_{i=1}^n \sum_{j=1}^n m_{ij}(t, \mathbf{q}) \dot{q}_i \dot{q}_j - T + V \\
&= T_1 + 2T_2 - (T_0 + T_1 + T_2) + V \\
&= T_0 + T_2 + V
\end{aligned}$$

It is apparent from the above expression that $T_0 = T_1 = 0$ is a sufficient condition that guarantees the Hamiltonian $\mathcal{H}$ equates to the total energy $T + V$. In other words, the system kinetic energy is a symmetric quadratic form in the generalized coordinate's time derivatives

$$T = T_2 = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n m_{ij}(t, \mathbf{q}) \dot{q}_i \dot{q}_j = \frac{1}{2} \dot{\mathbf{q}}^T [\mathbf{M}] \dot{\mathbf{q}} \quad (7.17)$$

Such a system is referred to as a **natural system**, which has a purely quadratic kinetic energy. For a natural system, T_0 usually comes from frame motion or motion constraints.

Example 7.1: Pendulum attached to moving cart is not a natural system.



An ideal pendulum is attached with the movable pivot (the cart is massless). The pivot motion be any prescribed function of time

$$\begin{aligned}
x(t) &= A \sin(\Omega t) \\
\dot{x}(t) &= A\Omega \cos(\Omega t) \\
\ddot{x}(t) &= -A\Omega^2 \sin(\Omega t)
\end{aligned}$$

Kinematics:

$$\begin{aligned}
\mathbf{R} &= A \sin(\Omega t) \hat{\mathbf{n}}_1 + r \hat{\mathbf{e}}_r \\
\dot{\mathbf{R}} &= A\Omega \cos(\Omega t) \hat{\mathbf{n}}_1 + r\dot{\theta} \hat{\mathbf{e}}_\theta
\end{aligned}$$

$$T = \frac{1}{2} m \dot{\mathbf{R}} \cdot \dot{\mathbf{R}} = \frac{1}{2} m \left[(A\Omega \cos \Omega t)^2 + r^2 \dot{\theta}^2 + 2r\dot{\theta}(A\Omega \cos \Omega t) \cos \theta \right]$$

Therefore, we have the three parts of T as:

$$T_0 = \frac{1}{2} m A^2 \Omega^2 \cos^2(\Omega t)$$

$$T_1 = m A \Omega r \cos \theta \cos(\Omega t) \dot{\theta}$$

$$T_2 = \frac{1}{2} m r^2 \dot{\theta}^2$$

Relationship of $\mathcal{H}$ to Work/Energy Integral

Integrate the work of non-conservative forces and using the Lagrange's equations for non-conservative forces, the relationship of the change of the total energy and the change of Hamiltonian $\mathcal{H}$ is found as (detailed derivations referred to textbook):

$$\frac{d}{dt}(T + V) = \frac{d\mathcal{H}}{dt} + \sum_{i=1}^N \mathbf{f}_{nc_i} \cdot \frac{\partial \mathbf{R}_i}{\partial t} + \frac{\partial \mathcal{L}}{\partial t} \equiv \sum_{i=1}^N \mathbf{f}_{nc_i} \cdot \dot{\mathbf{R}}_i \quad (7.28)$$

or in the integral format:

$$(T + V) \Big|_{t_0}^{t_f} - \mathcal{H} \Big|_{t_0}^{t_f} = \int_{t_0}^{t_f} \sum_{i=1}^N \mathbf{f}_{nc_i} \cdot \frac{\partial \mathbf{R}_i}{\partial \tau} d\tau + \int_{t_0}^{t_f} \frac{\partial \mathcal{L}}{\partial \tau} d\tau = \int_{t_0}^{t_f} \sum_{i=1}^N \mathbf{f}_{nc_i} \cdot \dot{\mathbf{R}}_i d\tau \quad (7.29)$$

If the Lagrangian $\mathcal{L}$ does not explicitly depend on time, then the Hamiltonian $\mathcal{H}$ is identical to the total energy.

Hamilton's Canonical Equations (Canonical Form)

The Lagrangian function $\mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t)$ is always expressed as a function of generalized coordinate q_i and generalized velocity $\dot{q}_i$, but never the generalized momentum p_i .

Differently, Hamiltonian can be expressed in two formats:

$$\mathcal{H} \equiv \sum_{j=1}^n \frac{\partial \mathcal{L}}{\partial \dot{q}_j} \dot{q}_j - \mathcal{L}(t, \mathbf{q}, \dot{\mathbf{q}}) \equiv F(\mathbf{q}, \dot{\mathbf{q}}, t) \quad (7.52)$$

or

$$\mathcal{H} = \sum_{j=1}^n \mathbf{p}_j g_j(\mathbf{q}, \mathbf{p}, t) - \mathcal{L}(\mathbf{q}, \mathbf{g}(\mathbf{q}, \mathbf{p}, t), t) \equiv G(\mathbf{q}, \mathbf{p}, t) \quad (7.53)$$

where $g_j(\mathbf{q}, \mathbf{p}, t) = \dot{q}_j$ and $f_j(\mathbf{q}, \dot{\mathbf{q}}, t) = p_j$ are the two transformations.

This distinction is important because when we calculate partial derivatives, $\frac{\partial \mathcal{H}}{\partial q_i}$ will have different results for the above two expressions $F(\mathbf{q}, \dot{\mathbf{q}}, t)$ and $G(\mathbf{q}, \mathbf{p}, t)$. To prevent confusion when there is need to be specific, we introduce a notation for differentiation of an arbitrary function $\mathcal{F}$

$$\left[\frac{\partial \mathcal{F}}{\partial q_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} \quad \text{when } (\mathbf{q}, \dot{\mathbf{q}}, t) \text{ is used} \quad (7.56)$$

$$\left[\frac{\partial \mathcal{F}}{\partial q_j} \right]_{\mathbf{q}, \mathbf{p}} \quad \text{when } (\mathbf{q}, \mathbf{p}, t) \text{ is used} \quad (7.57)$$

Lagrange's equations can be re-written using these new notations as

$$\dot{p}_j = \left[\frac{\partial \mathcal{L}}{\partial q_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} + Q_{nc_j} \quad (7.60)$$

$$\begin{aligned} \left[\frac{\partial \mathcal{H}}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} &= \left[\frac{\partial}{\partial p_k} \left(\sum_{j=1}^n p_j \dot{q}_j - \mathcal{L}(\mathbf{q}, \dot{\mathbf{q}}, t) \right) \right]_{\mathbf{q}, \mathbf{p}} \\ &= \dot{q}_k + \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} - \left\{ \sum_{j=1}^n \left[\frac{\partial \mathcal{L}}{\partial q_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} \left[\frac{\partial q_j}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} + \sum_{j=1}^n \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} \left[\frac{\partial \dot{q}_j}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} \right\} \\ &= \dot{q}_k + \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} - \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial p_k} \right]_{\mathbf{q}, \mathbf{p}} \\ &= \dot{q}_k \end{aligned} \quad (7.61)$$

Using similar tricks, we get

$$\begin{aligned}
\left[\frac{\partial \mathcal{H}}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} &= \left[\frac{\partial}{\partial q_k} \left(\sum_{j=1}^n p_j \dot{q}_j - \mathcal{L} \right) \right]_{\mathbf{q}, \mathbf{p}} \\
&= \sum_{j=1}^n \left[\frac{\partial p_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} q_j + \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} - \left[\frac{\partial \mathcal{L}}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} \\
&= \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} - \left(\sum_{j=1}^n \left[\frac{\partial \mathcal{L}}{\partial q_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} \left[\frac{\partial q_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} + \sum_{j=1}^n \left[\frac{\partial \mathcal{L}}{\partial \dot{q}_j} \right]_{\mathbf{q}, \dot{\mathbf{q}}} \left[\frac{\partial \dot{q}_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} \right) \\
&= \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} - \left[\frac{\partial \mathcal{L}}{\partial q_k} \right]_{\mathbf{q}, \dot{\mathbf{q}}} - \sum_{j=1}^n p_j \left[\frac{\partial \dot{q}_j}{\partial q_k} \right]_{\mathbf{q}, \mathbf{p}} \\
&= - \left[\frac{\partial \mathcal{L}}{\partial q_k} \right]_{\mathbf{q}, \dot{\mathbf{q}}}
\end{aligned} \tag{7.64}$$

Finally, the [Hamilton's canonical equations](#) are given as

$$\begin{aligned}
\dot{q}_j &= \left[\frac{\partial \mathcal{H}}{\partial p_j} \right]_{\mathbf{q}, \mathbf{p}} \\
\dot{p}_j &= - \left[\frac{\partial \mathcal{H}}{\partial q_j} \right]_{\mathbf{q}, \mathbf{p}} + Q_{nc_j}
\end{aligned} \tag{7.65}$$

for $j = 1, 2, \dots, n$. The explicit notation on the independent set of coordinates used for the partial derivatives is dropped, with the implicit assumption that $(\mathbf{q}, \mathbf{p}, t)$ are chosen as the independent variables for partial derivatives.

Introducing the canonical state vector

$$\mathbf{x} = \begin{bmatrix} \mathbf{q} \\ \mathbf{p} \end{bmatrix}$$

then the gradient vector is

$$\frac{\partial \mathcal{H}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial \mathcal{H}}{\partial \mathbf{q}} \\ \frac{\partial \mathcal{H}}{\partial \mathbf{p}} \end{bmatrix}$$

and we have the matrix form of Hamilton's equations as (todo: verify if the negative sign is correct)

$$\dot{\mathbf{x}} = -[J] \frac{\partial \mathcal{H}}{\partial \mathbf{x}} + \begin{bmatrix} \mathbf{0} \\ \mathbf{Q}_{nc} \end{bmatrix}, \tag{7.66}$$

where the skew-symmetric matrix $[J]$ is

$$[J] = \begin{bmatrix} 0_n & -I_n \\ I_n & 0_n \end{bmatrix} \tag{7.69}$$

The matrix $[J]$ is orthogonal as

$$[J]^T \cdot [J] = [I_{2n}]$$

and also has a special feature that

$$[J] \cdot [J] = -[I_{2n}]$$

The matrix $[J]$ belongs to a larger group of special matrix, the symplectic matrix.

🔗 $[J]$ is very much like an active rotation matrix for 90 degrees.

Canonical Coordinate Transformations

Suppose $(\mathbf{q}, \mathbf{p})$ are a set of canonical variables associated with a Hamiltonian $\mathcal{H}(\mathbf{q}, \mathbf{p}, t)$ and these variables satisfy a set of differential equations of the form of Eqs. (7.66), given a general smooth differentiable set of $2n$ transformation functions:

$$\begin{aligned}
q_j &= f_j(\mathbf{q}^*, \mathbf{p}^*, t) \\
p_j &= g_j(\mathbf{q}^*, \mathbf{p}^*, t)
\end{aligned} \tag{7.78}$$

and an associated smooth function $\mathcal{H}^*$. We will find the conditions to guarantee the same canonical structure.

Lagrange bracket conditions (fundamental one)

Take time derivatives of the above transformations:

$$\begin{aligned}\frac{dq_j}{dt} &= \frac{\partial f_j}{\partial t} + \sum_{i=1}^n \left(\frac{\partial f_j}{\partial q_i^*} \frac{\partial q_i^*}{\partial t} + \frac{\partial f_j}{\partial p_i^*} \frac{\partial p_i^*}{\partial t} \right) = \frac{\partial \mathcal{H}}{\partial p_j} \\ \frac{dp_j}{dt} &= \frac{\partial g_j}{\partial t} + \sum_{i=1}^n \left(\frac{\partial g_j}{\partial q_i^*} \frac{\partial q_i^*}{\partial t} + \frac{\partial g_j}{\partial p_i^*} \frac{\partial p_i^*}{\partial t} \right) = -\frac{\partial \mathcal{H}}{\partial q_j}\end{aligned}$$

where we have assumed a conservative system to get rid of $\mathbf{Q}$ for simplicity. It can be added back later without affecting the analysis here.

For the partial derivative of any new coordinate α picked from $\mathbf{q}$ or $\mathbf{p}$

$$\begin{aligned}\frac{\partial \mathcal{H}}{\partial \alpha} &= \sum_{j=1}^n \left(\frac{\partial \mathcal{H}}{\partial p_j} \frac{\partial p_j}{\partial \alpha} + \frac{\partial \mathcal{H}}{\partial q_j} \frac{\partial q_j}{\partial \alpha} \right) \\ &= \sum_{j=1}^n \left(\frac{\partial \mathcal{H}}{\partial p_j} \frac{\partial g_j}{\partial \alpha} + \frac{\partial \mathcal{H}}{\partial q_j} \frac{\partial f_j}{\partial \alpha} \right) \\ &= \sum_{j=1}^n \left(\frac{\partial f_j}{\partial t} \frac{\partial g_j}{\partial \alpha} - \frac{\partial g_j}{\partial t} \frac{\partial f_j}{\partial \alpha} \right) \\ &\quad + \sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial f_j}{\partial q_i^*} \frac{\partial q_i^*}{\partial t} \frac{\partial g_j}{\partial \alpha} - \frac{\partial g_j}{\partial q_i^*} \frac{\partial q_i^*}{\partial t} \frac{\partial f_j}{\partial \alpha} \right) \\ &\quad + \sum_{i=1}^n \sum_{j=1}^n \left(\frac{\partial f_j}{\partial p_i^*} \frac{\partial p_i^*}{\partial t} \frac{\partial g_j}{\partial \alpha} - \frac{\partial g_j}{\partial p_i^*} \frac{\partial p_i^*}{\partial t} \frac{\partial f_j}{\partial \alpha} \right) \\ &= [\alpha, \beta] + \sum_{i=1}^n [q_i^*, \alpha] \frac{dq_i^*}{dt} + \sum_{i=1}^n [p_i^*, \alpha] \frac{dp_i^*}{dt}\end{aligned} \tag{7.87 and 7.89}$$

where we have used the definition of **Lagrange's bracket** notation given by

$$[* , \alpha] = \sum_{j=1}^n \left(\frac{\partial f_j}{\partial * } \frac{\partial g_j}{\partial \alpha} - \frac{\partial g_j}{\partial * } \frac{\partial f_j}{\partial \alpha} \right) \tag{7.88 altered}$$

So, sufficient conditions to guarantee a canonical transformation are the following skew-symmetric conditions on Lagrange's brackets ($k \neq m$):

$$\begin{aligned}[t, \alpha] &= 0 \\ [q_k^*, q_m^*] &= 0 \\ [p_k^*, p_m^*] &= 0 \\ [q_k^*, p_m^*] &= 0 \\ [q_k^*, p_k^*] &= +1 \\ [p_k^*, q_k^*] &= -1\end{aligned} \tag{7.90}$$

The canonical transformation sufficiency conditions of Eqs. (7.90) and (7.91) hold for the case of general forces, with the generalized forces transformed using Eq. (7.93) below:

$$\begin{aligned}\mathcal{Q}_{q_j}^* &= \sum_{k=1}^n \left(\frac{\partial q_k}{\partial p_j^*} \mathcal{Q}_{q_k} - \frac{\partial f_k}{\partial p_j^*} \mathcal{Q}_{p_k} \right) \\ \mathcal{Q}_{p_j}^* &= \sum_{k=1}^n \left(\frac{\partial q_k}{\partial q_j^*} \mathcal{Q}_{q_k} - \frac{\partial f_k}{\partial q_j^*} \mathcal{Q}_{p_k} \right)\end{aligned}$$

Perfect Differential Criterion for Canonical Transformations (convenient when forward transformation available)

The perfect differential criterion is important because it is much more convenient to test than the Lagrange bracket conditions.

Assume the transformation is given by

$$\begin{aligned}q_j &= q_j(q_1^*, \dots, q_n^*; p_1^*, \dots, p_n^*) \\ p_j &= p_j(q_1^*, \dots, q_n^*; p_1^*, \dots, p_n^*)\end{aligned} \tag{7.94}$$

Theorem 7.1: The transformation of Eq. (7.94) leads to q_j^* and p_j^* that satisfy canonical differential equations if the following perfect differential condition is true:

$$\sum_{j=1}^n (q_j^* dp_j^* - q_j dp_j) - d\mathcal{W} = 0 \tag{7.95}$$

for $\mathcal{W}$ being an arbitrarily smooth differentiable function. The new Hamiltonian $\mathcal{H}^*$ is obtained by simply replacing q_j and p_j by Eq. (7.94) in the original Hamiltonian.

Proof Sketch: We need to verify all the **Lagrange bracket conditions** in the previous section.

The essential is to find two expressions for $\frac{\partial^2 \mathcal{W}}{\partial q_k^* \partial q_l^*}, \frac{\partial^2 \mathcal{W}}{\partial q_k^* \partial p_l^*}, \frac{\partial^2 \mathcal{W}}{\partial p_k^* \partial p_l^*}$, which will be then used to formulas desired Lagrange brackets.

Specifically, by plugging in the time derivative differentiating of the smooth transformation function p_i

$$dp_j = \sum_{k=1}^n \left(\frac{\partial p_j}{\partial q_k^*} dq_k^* + \frac{\partial p_j}{\partial p_k^*} dp_k^* \right)$$

into the time derivative the smooth function $\mathcal{W}$

$$d\mathcal{W} = \sum_{k=1}^n \left(\frac{\partial \mathcal{W}}{\partial q_k^*} dq_k^* + \frac{\partial \mathcal{W}}{\partial p_k^*} dp_k^* \right),$$

we obtain the equation after collecting terms in regard of dq_k^* and dp_k^*

$$\sum_{k=1}^n \left(\frac{\partial \mathcal{W}}{\partial q_k^*} dq_k^* + \frac{\partial \mathcal{W}}{\partial p_k^*} dp_k^* \right) = \sum_{k=1}^n \left(q_k^* - \sum_{j=1}^n q_j \frac{\partial p_j}{\partial p_k^*} \right) dp_k^* - \sum_{k=1}^n \sum_{j=1}^n q_j \frac{\partial p_j}{\partial q_k^*} dq_k^*.$$

Equating the coefficients and we get

$$\begin{aligned} \frac{\partial \mathcal{W}}{\partial q_k^*} &= - \sum_{j=1}^n q_j \frac{\partial p_j}{\partial q_k^*} \\ \frac{\partial \mathcal{W}}{\partial p_k^*} &= \left(q_k^* - \sum_{j=1}^n q_j \frac{\partial p_j}{\partial p_k^*} \right) \end{aligned}$$

which eventually leads to all the second order partial differentiations we need.

The last step is to verify Lagrange bracket conditions in Eq. (7.90) one by one.

Example 7.6: An orthogonal transformation is canonical.

Consider the orthogonal transformation

$$\mathbf{q} = [A] \mathbf{q}^* \quad \mathbf{p} = [A] \mathbf{p}^*$$

where $[A]$ is an orthogonal matrix.

It can be easily verified that $\mathbf{q}^{*T} d\mathbf{p}^* - \mathbf{q}^T d\mathbf{p} = \mathbf{q}^{*T} d\mathbf{p}^* - ([A] \mathbf{q}^*)^T [A] \mathbf{p}^*$

$$= \mathbf{q}^{*T} d\mathbf{p}^* - \mathbf{q}^{*T} [A]^T [A] \mathbf{p}^* = 0 = d\mathcal{W}$$

where $\mathcal{W} = 0$ is a smooth differentiable function.

Transformation Jacobian Perspective on Canonical Transformations (convenient when inverse transformation available)

Let the generalized coordinate and momenta variables be written as

$$\begin{aligned} (q_1, \dots, q_n; p_1, \dots, p_n) &\equiv (x_1, \dots, x_n, x_{n+1}, \dots, x_{2n}) \\ (q_1^*, \dots, q_n^*; p_1^*, \dots, p_n^*) &\equiv (X_1, \dots, X_n, X_{n+1}, \dots, X_{2n}) \end{aligned}$$

Consider the transformation

$$\mathbf{x} = \mathbf{F}(\mathbf{X})$$

The inverse transformation is denoted as

$$\mathbf{X} = \mathbf{G}(\mathbf{x})$$

and its Jacobian matrix $[M]$ is

$$[M] = \left[\frac{\partial \mathbf{G}}{\partial \mathbf{x}} \right] \equiv \left[\frac{\partial \mathbf{X}}{\partial \mathbf{x}} \right].$$

The conditions for a canonical transformation is the symplectic requirement for the Jacobian matrix $[M]$:

$$[M][J][M]^T = [J]$$

where $[J]$ is the skew symmetric and orthogonal matrix given as

$$[J] = \begin{bmatrix} 0_n & -I_n \\ I_n & 0_n \end{bmatrix}.$$

Proof Sketch: After expanding $[M]$ in elements, we will find the LHS contains all the Lagrange brackets.

$$[M][J][M]^T = \begin{bmatrix} [q_i^*, q_j^*] & [q_i^*, p_j^*] \\ [p_k^*, q_j^*] & [p_k^*, p_j^*] \end{bmatrix} = [J]$$

Applying the Lagrange bracket conditions and it becomes the symplectic matrix $[J]$.

Poisson's Brackets

Consider the time variation of a general smooth function $\mathcal{F}(\mathbf{q}, \mathbf{p}, t)$, we define the **Poisson's brackets** of $\mathcal{F}$ and $\mathcal{H}$ as

$$(\mathcal{F}, \mathcal{H}) \equiv \sum_{i=1}^n \left(\frac{\partial \mathcal{F}}{\partial q_i} \frac{\partial \mathcal{H}}{\partial p_i} - \frac{\partial \mathcal{F}}{\partial p_i} \frac{\partial \mathcal{H}}{\partial q_i} \right)$$

or in the matrix format

$$(\mathcal{F}, \mathcal{H}) \equiv \left[\frac{\partial \mathcal{F}}{\partial \mathbf{q}} \right]^T \left[\frac{\partial \mathcal{H}}{\partial \mathbf{p}} \right] - \left[\frac{\partial \mathcal{F}}{\partial \mathbf{p}} \right]^T \left[\frac{\partial \mathcal{H}}{\partial \mathbf{q}} \right]$$

The derivative of $\mathcal{F}$ is

$$\begin{aligned} \frac{d\mathcal{F}}{dt} &= \left[\frac{\partial \mathcal{F}}{\partial \mathbf{q}} \right]^T \dot{\mathbf{q}} + \left[\frac{\partial \mathcal{F}}{\partial \mathbf{p}} \right]^T \dot{\mathbf{p}} + \frac{\partial \mathcal{F}}{\partial t} \\ &= \left[\frac{\partial \mathcal{F}}{\partial \mathbf{q}} \right]^T \left[\frac{\partial \mathcal{H}}{\partial \mathbf{p}} \right] - \left[\frac{\partial \mathcal{F}}{\partial \mathbf{p}} \right]^T \left[\frac{\partial \mathcal{H}}{\partial \mathbf{q}} \right] + \left[\frac{\partial \mathcal{F}}{\partial \mathbf{p}} \right]^T \mathbf{Q}_{nc} + \frac{\partial \mathcal{F}}{\partial t} \\ &= (\mathcal{F}, \mathcal{H}) + \left[\frac{\partial \mathcal{F}}{\partial \mathbf{p}} \right]^T \mathbf{Q}_{nc} + \frac{\partial \mathcal{F}}{\partial t} \end{aligned} \quad (7.73 \text{ and } 7.74)$$

Poisson's brackets reveal more algebraic structure of Hamiltonian dynamics. This will be used frequently in courses link geometric mechanics.

📌 Poisson brackets are more commonly written as $\{\mathcal{F}, \mathcal{H}\}$ in sources outside of our textbook. However, differences in notation conventions should not be seen as a barrier to understanding.

Summary

This chapter is just a brief introduction to basics of Hamiltonian canonical forms and transformations. The richer content about Hamiltonian systems and dynamical system theories to analyzing the system behavior are beyond the scope of this course.

- Hamiltonian $\mathcal{H}$ is not total energy generally.
- Hamiltonian $\mathcal{H}$ is not constant generally.
- has units of energy
- is frequently a constant of the motion, even for a large class of nonconservative systems
- is sometimes but not always, equal to the total energy of the system.